

Varun Teja Kulla

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Summary

Architect and deliver multi-agent AI solutions using LangChain, LangGraph, and OpenAI LLM ecosystems. Engineer scalable AI backend platforms leveraging FastAPI, Redis, Docker, and microservice architectures. Orchestrate ETL and MLOps pipelines using Apache Airflow, MLflow, and CI/CD workflows to streamline model lifecycle management. Accelerate LLM inference performance using vLLM through model fine-tuning and quantization strategies. Leverage computer vision and deep learning methodologies to power real-time AI-driven applications.

Education

Centurion University

July 2022 - March 2025

Bachelor of Science in Computer Science and Artificial Intelligence (GPA: 8.8 / 10.0)

Vizianagaram, AP

- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Deep Learning, Computer Vision, Image Analytics, and Model Fine-tuning.

AIMS Engineering College

July 2019 - May 2022

Diploma in Electrical and Electronics Engineering (GPA: 8.9 / 10.0)

Mummidivaram, AP

- **Relevant Coursework:** Electric Motor Capacity, Power Systems, Digital Electronics, Electrical Monitoring, and Load Testing.

Experience

Finrapt Technologies Private Ltd

May 2025 – Present

GenAI Developer

Hyderabad, Telangana

- Architected a multi-agent transaction chatbot supporting multi-turn conversations, improving response accuracy and reducing context loss by 30%.
- Orchestrated end-to-end ETL pipelines using Apache Airflow and MLflow, reducing manual deployment effort by 40% and increasing training workflow reliability.
- Engineered CI/CD workflows for FastAPI model serving and optimized LLM inference using vLLM and Ollama, lowering latency by 25%.
- Coordinated agent orchestration using LangGraph and LlamaIndex while containerizing services with Docker to improve scalability and deployment consistency.
- Deployed and maintained self-hosted GitLab on Linux servers, managing repositories, runners, and access controls for streamlined CI/CD operations.
- Automated user provisioning and access management using Linux user/group policies and Windows Active Directory integration, reducing manual onboarding effort.

March 2025 – May 2025

Python Developer Intern

Hyderabad, Telangana

- Engineered a real-time face and identity verification platform using YOLOv8, boosting authentication accuracy by 20%.
- Refactored legacy Java modules into Python, decreasing technical debt and improving code maintainability by 35%.
- Constructed high-throughput FastAPI backend services supporting thousands of concurrent inference requests, strengthening system stability and throughput.
- Implemented automated user creation and authentication workflows integrating Linux systems with Active Directory, improving security and consistency in access management.

Technical Skills

Languages: Python, SQL, C, Java (Basic)

GenAI & Agentic AI: LangChain, LangGraph, LlamaIndex, OpenAI APIs, Prompt Engineering, Multi-Agent Systems, RAG Pipelines

Linux & System Administration: Systemd (systemctl, service management), Cron Jobs & Scheduling, Shell Scripting (Bash), User & Group Management, File Permissions & ACLs, SSH & Secure Access, Process Management, Package Management (APT/YUM), Log Management (journalctl), Server Configuration, GitLab Setup, System Monitoring & Troubleshooting

LLM Deployment & Optimization: vLLM, Ollama, Model Fine-Tuning, Quantization, Model Serving, Inference Optimization

Machine Learning & Computer Vision: PyTorch, TensorFlow, YOLOv8, Deep Learning, Image Processing, Model Training & Evaluation

Backend & Microservices: FastAPI, REST APIs, Uvicorn/Gunicorn, Microservice Architecture, Redis

MLOps & Data Engineering: Apache Airflow, MLflow, ETL Pipelines, Experiment Tracking, Model Lifecycle Management

DevOps & Deployment: Docker, CI/CD Pipelines, Containerization, API Deployment, Linux

Databases: PostgreSQL, Redis

Projects

Multi-Agent Transaction Chatbot System | *FastAPI, Redis, OpenAI, LangGraph, LangChain*

- Architected a multi-agent transaction chatbot leveraging LangGraph and LangChain to enable dynamic task orchestration and conversational workflow management.
- Integrated OpenAI LLM services with FastAPI backend infrastructure to deliver real-time conversational transaction support.
- Engineered Redis-backed memory management to preserve agent state, improving context retention and minimizing response inconsistencies.
- Created reusable prompt templates to standardize agent instructions and enhance response consistency and scalability.

Automated ETL and Model Training Pipeline | *Apache Airflow, MLflow, Python, Microservices, Docker*

- Orchestrated a scheduled ETL workflow using Apache Airflow to aggregate data from distributed microservices at hourly intervals.
- Engineered preprocessing and data validation pipelines to improve dataset quality for downstream machine learning workflows.
- Established an end-to-end ML training and experiment tracking framework using MLflow for versioning and performance monitoring.

Face and Identity Verification System | *YOLOv8, FastAPI, Computer Vision, PostgreSQL, Python*

- Engineered a real-time face and identity authentication pipeline using YOLOv8 and computer vision techniques.
- Constructed FastAPI microservices to manage credential validation and inference orchestration workflows.
- Established PostgreSQL-backed identity storage for secure metadata and verification record management.
- Optimized model training and evaluation workflows to improve detection accuracy and authentication reliability.

Real-Time Vehicle Detection and Tracking System | *YOLOv8, ONNX, Computer Vision, Python, OpenCV*

- Architected a real-time vehicle detection and tracking solution using YOLOv8 for traffic and surveillance analytics.
- Implemented multi-object tracking pipelines to analyze vehicle movement and generate structured tracking metadata.
- Converted YOLOv8 detection models into ONNX format to support lightweight inference and cross-platform edge deployment.
- Enhanced inference performance and reduced latency for deployment in resource-constrained environments.